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Error-controlled non-additive interaction discovery in machine learning models

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Machine learning (ML) models are powerful tools for detecting complex patterns, yet their 'black-box' nature limits their interpretability, hindering their use in critical domains like healthcare and finance. Interpretable ML methods aim to explain how features influence model predictions but often focus on univariate feature importance, overlooking complex feature interactions. Although recent efforts extend interpretability to feature interactions, existing approaches struggle with robustness and error control, especially under data perturbations. In this study, we introduce Diamond, a method for trustworthy feature interaction discovery. Diamond uniquely integrates the model-X knockoffs framework to control the false discovery rate, ensuring a low proportion of falsely detected interactions. Diamond includes a non-additivity distillation procedure that refines existing interaction importance measures to isolate non-additive interaction effects and preserve false discovery rate control. This approach addresses the limitations of off-the-shelf interaction measures, which, when used naively, can lead to inaccurate discoveries. Diamond's applicability spans a broad class of ML models, including deep neural networks, transformers, tree-based models and factorization-based models. Empirical evaluations on both simulated and real datasets across various biomedical studies demonstrate its utility in enabling reliable data-driven scientific discoveries. Diamond represents a significant step forward in leveraging ML for scientific innovation and hypothesis generation.

Machine learning (ML) has emerged as a critical tool in many application domains, largely due to its ability to detect subtle relationships and patterns within complex data[1]. Although the complexity of ML models contributes to their power, it also makes them challenging to interpret, leaving users with few clues about the mechanisms underlying a given model's outputs. Consequently, this black-box nature of ML models, particularly deep neural networks (DNNs), has hindered their applicability in error-intolerant domains like healthcare and finance. Stakeholders such as clinicians need to understand why and how the models make predictions before making important decisions, such as disease diagnosis[2]. Importantly, without providing insight into their internal mechanisms, ML models cannot be effectively used for making data-driven scientific discoveries, which are crucial for gaining human-understandable insights and driving successful innovation[3].

To enhance the interpretability of ML models for better data-driven scientific discoveries, interpretable ML methods have been developed to elucidate the internal mechanisms of these models[4]. These methods help to elucidate how individual features influence prediction

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outcomes by assigning an importance score to each feature so that higher scores indicate greater relevance to the prediction[5–9]. However, these univariate interpretations overlook a primary advantage of ML models: their ability to model complex interactions between features in a data-driven way. In fact, input features usually do not work individually within an ML model but cooperate with other features to make inferences jointly[10]. For example, it is well established in biology that genes do not operate in isolation but work together in co-regulated pathways with additive, cooperative or competitive interactions[11]. Additionally, gene–gene, gene–disease, gene–drug and gene–environment interactions are critical in explaining genetic mechanisms, diseases and drug effects[12].

Recognizing the limitations of univariate interpretations, efforts have been made to extend interpretable ML methods to discover interactions among features. These methods attribute the prediction influence to feature pairs and then rank candidate feature pairs from a trained ML model, with highly ranked pairs indicating higher importance[10,13–20]. However, it is important to note that these approaches characterize feature pairs in which both features are individually important for a model's prediction, rather than capturing the synergistic or interaction effects between the two features[21] (see the detailed explanation in the 'Measuring the non-additive interaction effect' section). Furthermore, the induced ranked list of feature pairs must be cut off at a certain confidence level for use in scientific discovery and hypothesis validation[10]. However, selecting this threshold is typically under user control and must be set arbitrarily. Worse still, existing methods are sensitive to perturbations, in the sense that even imperceivable perturbations of the input data may lead to dramatic changes in the importance ranking[22–24].

From a practitioner's perspective, a given set of discovered feature interactions is scientifically valuable only if a systematic strategy exists to prioritize and select relevant interactions in a robust and error-controlled manner, even in the presence of noise. Although many methods have been developed for feature interaction discovery, we are not aware of any previous attempts to conduct discovery and explicitly estimating and controlling the discovery error. Without this, accurate and reliable findings cannot be achieved. In this study, we introduce an error-controlled interaction discovery method for discovering interactions in ML models with a controlled error rate, named Diamond. Here the error is quantified by the false discovery rate (FDR)[25], which informally represents the expected proportion of falsely discovered interactions among all discovered interactions. A false discovery is a feature interaction that is discovered but not truly relevant.

Diamond consists of three key components (Fig. 1). First, Diamond achieves FDR control by leveraging the model-X knockoffs framework[26,27]. The core idea of this framework is to generate dummy features that perfectly mimic the empirical dependence structure among the original features and being conditionally independent of the response given the original features. Second, we discover that naively using off-the-shelf feature interaction importance measures cannot correctly control the FDR. To address this issue, we distil non-additive interaction effects from the interaction importance measures reported by existing methods, ensuring FDR control at the target level. Third, Diamond is applicable to a broad class of ML models, including DNNs, Kolmogorov–Arnold networks (KANs)[28], transformer models[29], tree-based models and factorization-based models. We have applied Diamond to various simulated and real datasets to demonstrate its empirical utility. Practically speaking, Diamond paves the way for the wider deployment of ML models in scientific discovery and hypothesis generation.

Results

Diamond overview

Consider a prediction task in which we have n independent and identically distributed samples $\mathbf{X} = \{x_i\}_{i=1}^n \in \mathbb{R}^{n\times p}$ and $\mathbf{Y} = \{y_i\}_{i=1}^n \in \mathbb{R}^{n\times 1}$, denoting the data matrix with p-dimensional features and the corresponding response, respectively. The response of the prediction task can be categorical labels such as disease status or numerical measurements such as body mass index (BMI), and the features could include gene expression data, microbial taxa abundance profiles and so on. The prediction task can be described using an ML model $f : \mathbb{R}^p \mapsto \mathbb{R}$ that maps from the input $x \in \mathbb{R}^p$ to the response $y \in \mathbb{R}$. When modelling the task, the function f learns the dependence structure of $\mathbf{Y}$ on $\mathbf{X}$, enabling effective prediction with the fitted ML model.

In this study, we focus on interpreting the fitted ML model for data-driven scientific discovery by identifying the non-additive feature interactions that contribute most to the model's predictions. We say that $\mathcal{I} \subset \{1 \dots p\}$ is a non-additive interaction of function f if and only if f cannot be decomposed into an addition of $|\mathcal{I}|$ subfunctions f_i, each of which excludes a corresponding interaction feature[10,30], that is, $f(x) \neq \sum_{i\in\mathcal{I}} f_i\left(x_{\{1,\dots,p\}\setminus i}\right)$. For example, the multiplication between two features x_i and x_j is a non-additive interaction because it cannot be decomposed into a sum of univariate functions, that is, $f(x_i, x_j) = x_i x_j \neq f_i(x_j) + f_j(x_i)$. On the other hand, $\log(x_i x_j)$ is considered to be an additive interaction because it can be decomposed in the logarithmic space, that is, $f(x_i, x_j) = \log(x_i x_j) = \log(x_j) + \log(x_i)$. Assume that there exists a group of interactions $\mathcal{S} = \{\mathcal{I}_1, \mathcal{I}_2, \dots\}$ such that, conditional on interactions $\mathcal{S}$, the response $\mathbf{Y}$ is independent of interactions in the complement $\mathcal{S}^c = \{1, \dots, p\} \times \{1, \dots, p\} \setminus \mathcal{S}$. In this setting, our goal is to accurately discover feature interactions in $\mathcal{S}$ without erroneously reporting too many incorrect interactions in $\mathcal{S}^c$.

Diamond is designed to discover feature interactions from fitted ML models and maintain a controlled FDR[25]. For a set of feature interactions $\hat{S} \subset \{1, \dots, p\} \times \{1, \dots, p\}$ discovered by some interaction detection method, the FDR is defined as

$$\mathrm{FDR} = \mathbb{E}[\mathrm{FDP}] \text{ with } \mathrm{FDP} = \frac{|\hat{S} \cap \mathcal{S}^c|}{|\hat{S}|}. \quad (1)$$

Commonly used procedures, such as the Benjamini–Hochberg procedure[25], achieve FDR control by working with p values computed against some null hypothesis. In the interaction discovery setting, for each feature interaction, one tests the significance of the statistical association between the specific interaction and the response, either jointly or marginally, and obtains a p value under the null hypothesis that the interaction is irrelevant. These p values are then used to rank the features for FDR control. However, controlling FDR in generic ML models, especially deep learning models, is challenging because, to our knowledge, the field lacks a method for producing meaningful p values that reflect interaction importance. To bypass the use of p values but still achieve FDR control, we draw inspiration from the model-X knockoffs framework[26,27]. The core idea of knockoffs is to generate dummy features that perfectly mimic the empirical dependence structure among the original features but are conditionally independent of the response given the original features. These knockoff features can then be used as a control by comparing the feature importance between the original features and their knockoff counterparts to estimate the FDR (Fig. 1a).

Diamond trains a generic ML model that takes as input an augmented data matrix $(\mathbf{X}, \tilde{\mathbf{X}}) \in \mathbb{R}^{n\times 2p}$, created by concatenating the data matrix $\mathbf{X} \in \mathbb{R}^{n\times p}$ with its knockoffs $\tilde{\mathbf{X}} \in \mathbb{R}^{n\times p}$ along the feature axis. After training, Diamond quantifies the feature interactions by interpreting the trained model and produces a ranked list of these interactions (Fig. 1b). However, existing feature importance measures cannot directly capture non-additive interactions. Geometrically, the marginal or interaction importance reported by current interpretation methods correspond to the projection of the total contribution to the prediction onto a one-dimensional feature axis or a two-dimensional feature–feature plane. Non-additive effects, however, represent the difference

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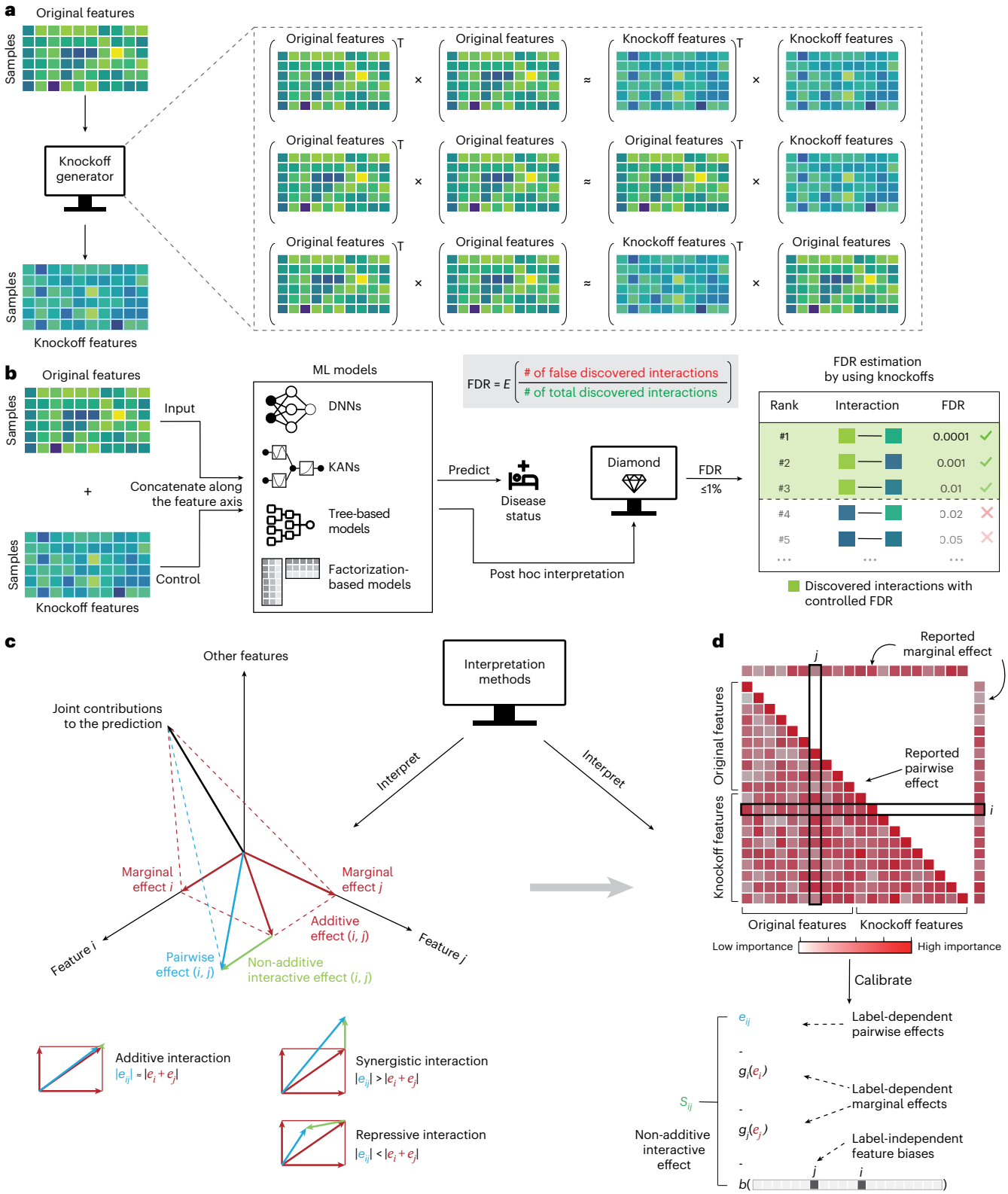


Fig. 1 | Overview of Diamond. a, Diamond achieves FDR control by leveraging knockoffs—dummy features that perfectly mimic the empirical dependence structure among the original features. **b**, Diamond trains generic ML models using both original features and their knockoff counterparts as inputs. Diamond quantifies feature interactions from trained ML models and produces a ranked list of these interactions with the estimated FDR, allowing users to confidently determine a cut-off threshold based on their desired confidence level. **c**, Existing feature importance measures are unable to directly capture non-additive interactions. Geometrically, the marginal or interaction importance reported by current interpretation methods correspond to the projection of the total contribution to the prediction onto a one-dimensional feature axis or a two-dimensional feature–feature plane. Non-additive effects, however, represent the difference between interaction effects and marginal effects. These non-additive effects can manifest either synergistic or repressive interactions. **d**, Diamond distils non-additive effects from the interaction importance measures reported by existing methods. Distillation removes both label-dependent marginal effects and label-independent feature biases from the reported feature interactions, leaving only the label-dependent non-additive interaction effects.

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a

ID	Simulation function
F_1	$\pi^{x_1x_2}\sqrt{2x_3} - \sin^{-1}(x_4) + \log(x_3 + x_5) - \frac{x_9}{x_{10}}\sqrt{\frac{x_7}{x_8}} - x_2x_7$
F_2	$\pi^{x_1x_2}\sqrt{2\lvert x_3\rvert} - \sin^{-1}(0.5x_4) + \log(\lvert x_3 + x_5\rvert + 1) - \frac{x_9}{1+\lvert x_{10}\rvert}\sqrt{\frac{x_7}{1+\lvert x_8\rvert}} - x_2x_7$
F_3	$\exp\lvert x_1 - x_2\rvert + \lvert x_2x_3\rvert - x_3^{2\lvert x_4\rvert} + \log(x_4^2 + x_5^2 + x_7^2 + x_8^2) + x_9 + \frac{1}{1+x_{10}^2}$
F_4	$\exp\lvert x_1 - x_2\rvert + \lvert x_2x_3\rvert - x_3^{2\lvert x_4\rvert} + (x_1x_4)^2 + \log(x_4^2 + x_5^2 + x_7^2 + x_8^2) + x_9 + \frac{1}{1+x_{10}^2}$
F_5	$\frac{1}{1+x_1^2+x_2^2+x_3^2} + \sqrt{\exp(x_4 + x_5)} + \lvert x_6 + x_7\rvert + x_8x_9x_{10}$
F_6	$\exp(\lvert x_1x_2\rvert + 1) - \exp(\lvert x_3 + x_4\rvert + 1) + \cos(x_5 + x_6 - x_8) + \sqrt{x_8^2 + x_9^2 + x_{10}^2}$
F_7	$(\arctan(x_1) + \arctan(x_2))^2 + \max(x_3x_4 + x_6, 0) - \frac{1}{1+(x_4x_5x_6x_7x_8)^2} + (\frac{\lvert x_7\rvert}{1+\lvert x_9\rvert})^5 + \sum_{i=1}^{10} x_i$
F_8	$x_1x_2 + 2^{x_3+x_5+x_6} + 2^{x_3+x_4+x_5+x_7} + \sin(x_7\sin(x_8 + x_9)) + \arccos(0.9x_{10})$
F_9	$\tanh(x_1x_2 + x_3x_4)\sqrt{\lvert x_5\rvert} + \exp(x_5 + x_6) + \log((x_6x_7x_8)^2 + 1) + x_9x_{10} + \frac{1}{1+\lvert x_{10}\rvert}$
F_{10}	$\sinh(x_1 + x_2) + \arccos(\tanh(x_3 + x_5 + x_7)) + \cos(x_4 + x_5) + \sec(x_7x_9)$

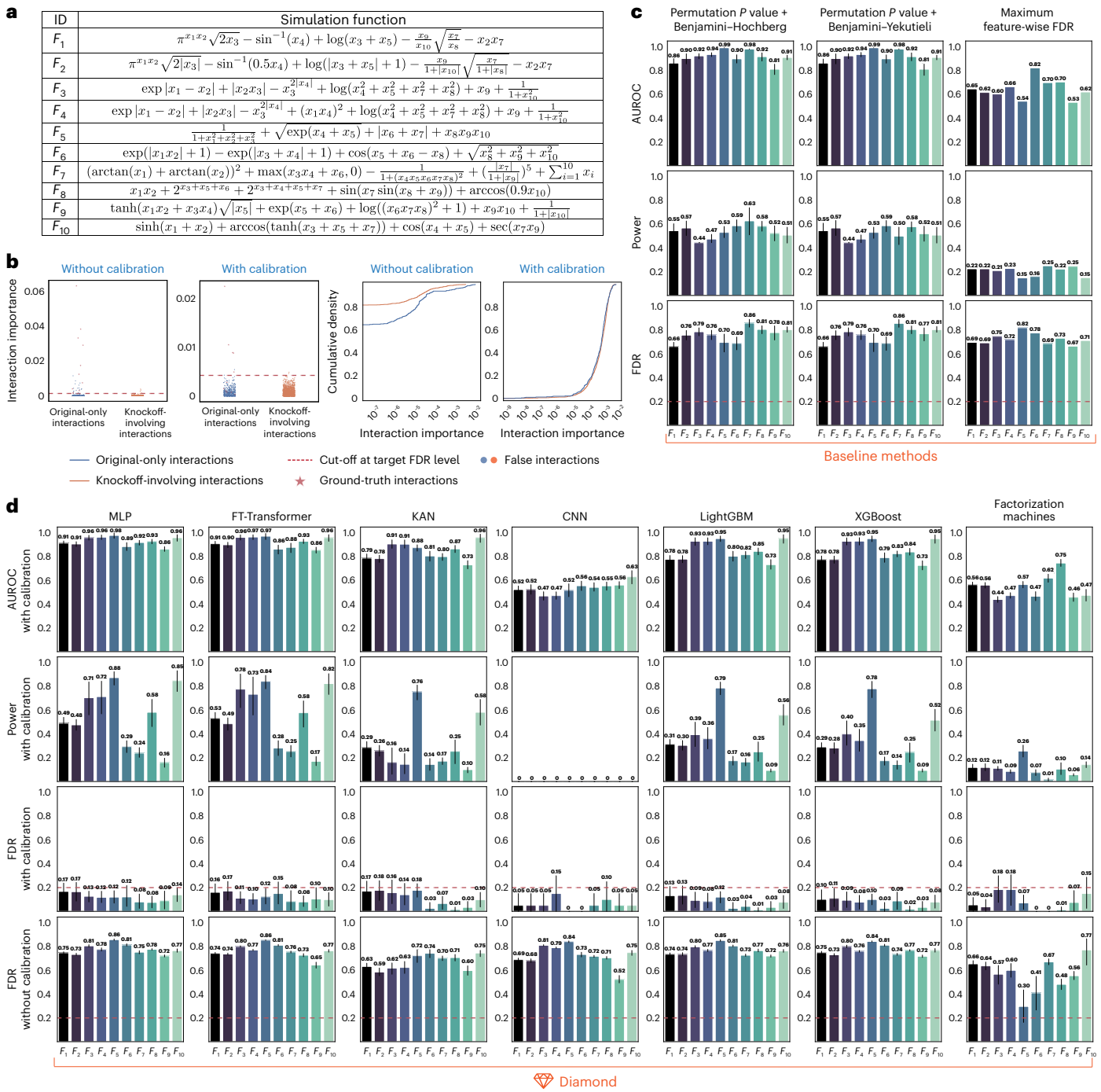


Fig. 2 | Evaluating Diamond for FDR control on simulated datasets. **a**, Evaluation based on a test suite of ten data-generating simulation functions proposed in ref. 10. **b**, Reported interaction importance from existing methods in simulation function F_1 reveals a clear distribution disparity between original-only interactions and those involving knockoffs. The distilled non-additive interactions help mitigate distributional disparities. **c**, Baseline methods fail to correctly control the FDR, thereby rendering the reported high power and AUROC invalid. **d**, Diamond identifies important non-additive interactions with controlled FDR, compatible with various ML models. The non-additivity distillation procedure is critical; without it, the FDR cannot be controlled. To ensure robustness, we repeated each experiment 20 times with different random seeds and reported the mean performance with 95% confidence intervals.

between the interaction effects and the marginal effects (Fig. 1c). Hence, Diamond distils non-additive interaction effects from the reported interaction importance measures from existing methods, thereby maintaining FDR control at the target level. The distillation process is designed to remove both label-dependent marginal effects and label-independent feature biases from the reported feature interactions, leaving only the label-dependent non-additive interaction effects (Fig. 1d).

Diamond discovers FDR-controlled interactions on simulated datasets

We started by evaluating the performance of Diamond on simulated datasets, assessing its ability to identify important non-additive interactions and controlling the FDR. We benchmarked Diamond on a test suite of ten simulated datasets generated by different simulation functions proposed in ref. 10. These datasets contain a mixture of univariate functions and multivariate interactions, exhibiting varied

order, strength and nonlinearity (Fig. 2a). Since our goal is to detect pairwise interactions, high-order interaction functions (for example, $F(x_1, x_2, x_3) = x_1x_2x_3$) are decomposed into pairwise interactions (for example, (x_1, x_2), (x_1, x_3), and (x_2, x_3)) to serve as the ground truth.

Following the settings used in ref. 10, we used a sample size of $n = 20{,}000$, equally divided into training and test sets. In addition, the number of features is set at $p = 30$, and all features are sampled randomly from a continuous uniform distribution, $U(0, 1)$. Only the first 10 out of 30 features contribute to the corresponding response, whereas the remaining features serve as noise to increase the task's complexity. For robustness, we repeated the experiment 20 times for each simulated dataset using different random seeds. Each repetition involved data generation, knockoff generation using KnockoffsDiagnostics[31], ML model training and interaction-wise FDR estimation. For all simulation settings, we reported the mean performance with 95% confidence intervals, fixing the target FDR level at $q = 0.2$.

Our analysis shows that Diamond consistently identifies important non-additive interactions with controlled FDR across all ML models (Fig. 2d). Under controlled FDR, the multilayer perceptron (MLP) and feature tokenizer transformer (FT-Transformer) models[29] demonstrate better performance in identifying important interactions than other ML models, as measured by the statistical power and the area under the receiver operating characteristic curve (AUROC). This superior performance is attributed to the inclusion of a knockoff-tailored plug-in pairwise-coupling layer design[32], which has been shown to maximize statistical power (Supplementary Section 1). It is important to note that certain ML models, such as convolutional neural networks, are not well suited for modelling tabular data due to their design for capturing local patterns in images or text. Even in the presence of model misspecification, Diamond maintains FDR control, although with a loss of power. This highlights Diamond's robustness and its broad applicability across various ML models.

We discovered that the proposed non-additivity distillation (see the 'Measuring the non-additive interaction effect' section) is critical; without it, the FDR cannot be controlled by naively using reported interaction importance values from existing methods (Fig. 2d and Supplementary Fig. 2). To gain insights into the FDR control failure in the absence of non-additivity distillation, we conducted a qualitative comparison assessing the interaction importance before and after distillation using the simulation function F_1 (Fig. 2b and Supplementary Fig. 3). The primary cause of the FDR control failure appears to lie in the distributional disparity between original-only interactions (that is, interactions involving two features from the original feature set rather than knockoffs) and interactions involving knockoffs. This observation suggests a violation of the knockoff filter's assumption in controlling the FDR (see the 'Measuring the non-additive interaction effect' section). The proposed non-additivity distillation procedure mitigates the disparity by extracting non-additive interaction effects from the reported interaction importance measures, thereby enhancing the utility of knockoff-involving interactions as a negative control for FDR estimation.

Finally, we verified whether alternative baseline methods can accurately identify important non-additive interactions with controlled FDR (Fig. 2c). We compared three baseline methods for FDR estimation: two relying on permutation-based interaction-wise p values, combined with the Benjamini–Hochberg and Benjamini–Yekutieli procedures, and one that represents interaction-wise FDR as an aggregation of feature-wise FDR (see the 'Alternative FDR estimation methods' section). Our analysis reveals that none of these baseline methods effectively control the FDR. This greatly reduces the utility of these methods, despite their reported high power and AUROC.

Diamond finds FDR-controlled interactions in simulations

The design of Diamond incorporates several key components, including knockoff generation and interaction importance measures. To demonstrate the robustness of Diamond in FDR control, we conducted a control study in which we modified Diamond by replacing these two components with alternative solutions. Specifically, we considered two variants of Diamond using the MLP model. In the first setting, we replaced KnockoffsDiagnostics with three alternatives: KnockoffGAN[33], Deep knockoffs[34] or VAE knockoffs[35], and we kept the MLP model and the interaction importance measure (that is, Expected Hessian[36]) unchanged. In the second setting, we replaced the interaction importance measure, Expected Hessian, with two alternatives: Integrated Hessian[16] and a model-specific interaction importance measure derived from the MLP weights (Supplementary Section 2), and we kept the MLP model and the knockoff generation unchanged. For each variant, we applied Diamond to the test suite of ten simulated datasets using the same settings.

The results indicate that Diamond is robust to various knockoff designs and importance measures. In the first study, our analysis shows that Diamond consistently discovers important non-additive interactions with controlled FDR across various knockoff designs (Fig. 3a). Despite the methodological differences in knockoff generation between KnockoffsDiagnostics, KnockoffGAN, Deep knockoffs and VAE knockoffs, Diamond achieves comparable statistical power and AUROC in each case. This result demonstrates the Diamond's stability in identifying important interactions in knockoff-based FDR estimation. Furthermore, it is worth noting that Diamond is robust even when the knockoffs are poorly generated. Specifically, we generated invalid knockoffs by independently permuting each feature across samples, thereby violating the definition of knockoffs outlined in the 'FDR control with knockoffs' section. Our analysis demonstrates that even in this extreme scenario, Diamond maintains FDR control, although with reduced power (Fig. 3b).

In the second study, our analysis shows that Diamond robustly identifies FDR-controlled non-additive interactions across a range of interaction importance measures (Fig. 3c). Despite the methodological differences between Expected Hessian, Integrated Hessian and model-specific measures, which involve mechanistic disparities in elucidating the relationships between features and responses, Diamond achieves valid FDR control and maintains reasonably good statistical power and AUROC. This result demonstrates Diamond's flexibility in identifying interactions based on various importance measures.

Finally, it is worth mentioning that regardless of the knockoff generation and interaction importance measures we used, there is room for improvement in achieving better statistical power and maintaining controlled FDR. Diamond tends to be conservative by overestimating the FDR, suggesting that an improved FDR control procedure could potentially boost statistical power.

Diamond finds interactions explaining disease progression

We then evaluated the performance of Diamond on a real dataset[37], assessing its ability to identify important interactions in the progression of disease. We used a quantitative study with $n = 442$ diabetes patients, each characterized by ten standardized baseline features, including age, sex, BMI, average blood pressure and six blood serum measurements (Fig. 4a). The task is to construct an ML model to predict the response of interest, which is a quantitative measure of disease progression one year after the baseline. We considered different types of ML model: MLP, FT-Transformer, KAN, LightGBM, XGBoost and factorization machines (FMs). We assessed the trained ML model using fivefold cross-validation and selected the one with the best performance for model interpretation and interaction discovery. For robustness, we repeated each experiment 20 times using different random seeds. Each repetition involved knockoff generation, ML model training and interaction-wise FDR estimation.

Our analysis shows that Diamond identifies the important interaction between BMI and serum triglyceride level (STL) across three different ML models (Fig. 4c). We investigated the identified interaction via

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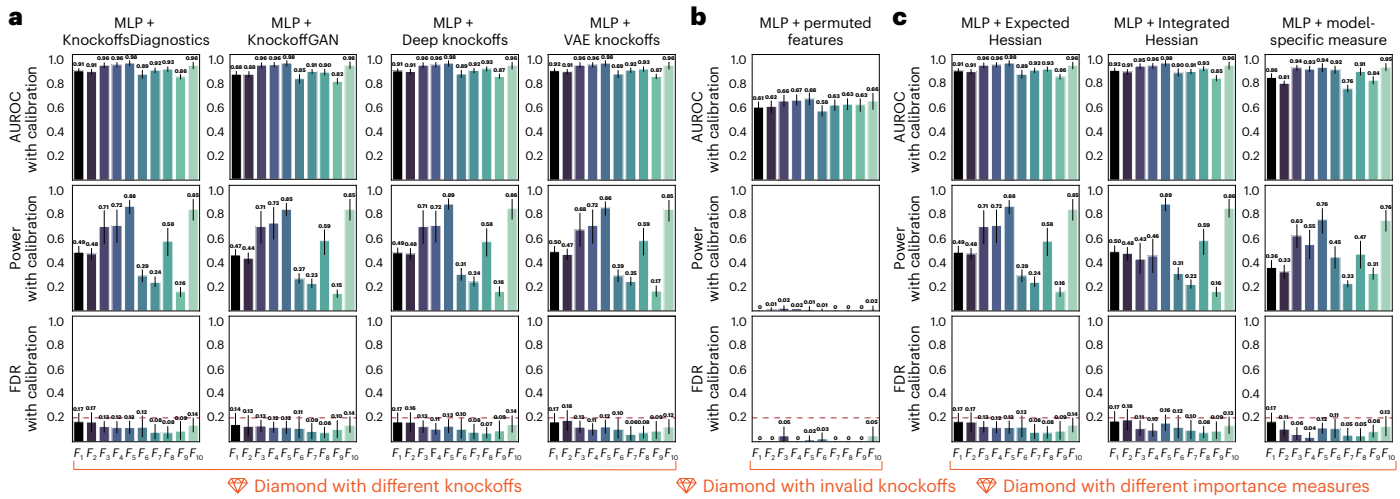


Fig. 3 | Evaluating Diamond for robustness on simulated datasets. **a**, Diamond demonstrates robustness across knockoffs generated by various methods, including KnockoffsDiagnostics, KnockoffGAN, Deep knockoffs and VAE knockoffs. **b**, Diamond maintains FDR control when paired with invalid knockoffs, generated by independently permuting each feature across samples, although at the cost of reduced power. **c**, Diamond demonstrates robustness across methodologically different interaction importance measures: Expected Hessian, Integrated Hessian and model-specific measures. To ensure robustness, we repeated each experiment 20 times with different random seeds and reported the mean performance with 95% confidence intervals.

literature search and found that high BMI and high STL are associated with an increased risk of diabetes, as reported by multiple studies[38–40]. The literature evidence is further supported by qualitative evaluation (Fig. 4d). Specifically, we examined the identified interactions from three perspectives: the marginal importance measure, the interaction importance measure and the contribution of the marginal importance measures to the interaction importance measure. This analysis showed that high BMI and high STL contribute to more severe progression of diabetes. Meanwhile, higher STL further synergistically reinforces the contribution of BMI to the progression of diabetes.

Unlike other ML models, FT-Transformer identifies two distinct interactions: the interaction between blood pressure and high-density lipoproteins, and the interaction between age and sex. We investigated the identified interactions through a literature search and found that high-density lipoprotein levels are positively associated with hypertension in individuals with elevated levels of circulating CD34-positive cells[41]. We also found that the prevalence of diabetes varies by sex, with men typically being diagnosed at a younger age and lower BMI, whereas young women with diabetes are currently less likely than men to receive treatment[42].

Finally, we find that Diamond does not necessarily report a lower FDR for interactions composed of two marginally important features. All ten baseline features from diabetes patients are effective in predicting disease progression, as measured by the Expected Gradient scores in the MLP model (Fig. 4a). Among these features, in addition to BMI and STL, blood pressure, blood sugar level and high-density lipoproteins are more predictive than others. It is important to note that Diamond still associates most interactions composed of these highly predictive features with a high FDR threshold.

Diamond probes enhancer activity in *Drosophila* embryos

We then evaluated the performance of Diamond on a *Drosophila* enhancer dataset[43] to investigate the relationship between enhancer activity and DNA occupancy for transcription factor (TF) binding and histone modifications in *Drosophila* embryos. We used a quantitative study of DNA occupancy for 23 TFs and 13 histone modifications with the enhancer status labelled for 7,809 genomic sequence samples from blastoderm *Drosophila* embryos (Fig. 4a). The enhancer status for each sequence is binarized as the response, depending on whether the sequence drives patterned expression in blastoderm embryos. To predict the enhancer status, the maximum value of normalized fold enrichment[44] of ChIP-seq and ChIP-chip assays for each TF or histone modification served as our features. We considered different types of ML model: MLP, FT-Transformer, KAN, LightGBM, XGBoost and FM. We assessed the trained ML model using fivefold cross-validation and selected the one with the best performance for model interpretation and interaction discovery. For robustness, we repeated the experiment 20 times using different random seeds. Each repetition involved knockoff generation, ML model training and interaction-wise FDR estimation.

We started by comparing the identified interactions against a list of well-characterized physical interactions in the early *Drosophila* embryos, collected in ref. 43. These interactions have been identified over decades and have been confirmed to play a critical role in regulating spatial and temporal patterning[45]. If Diamond is successful in identifying important interactions and controlling the FDR, then the identified interactions should considerably overlap with these previously reported physical interactions. Meanwhile, it is important to acknowledge that this list of interactions, proposed years ago, represents only a partial set of true interactions. Therefore, a detected interaction not appearing in the list does not necessarily indicate it is incorrect. Our results show that three of the top five interactions reported by Diamond are included in the ground-truth list for MLP and tree-based models, four out of five for FM, and all five for FT-Transformer (Fig. 5c). The only exception is KAN, which reported all but one interaction at the same FDR levels. In comparison, the baseline methods exhibit significantly different behaviours (Fig. 5b). Two baseline methods, which are based on permutation-based interaction-wise *p* values combined with the Benjamini–Hochberg and Benjamini–Yekutieli procedures, reported no ground-truth interactions among its top five identifications. Another method, which aggregates feature-wise FDR, reported ground-truth interactions along with a large number of non-ground-truth interactions.

Next, we investigated the identified interactions that are not included in the ground-truth list. For example, we investigated the interaction between the TFs Snail (UniProt ID: P08044) and Twist (UniProt ID: P10627) identified by Diamond+MLP through databases containing assorted experimentally verified interactions[46] and literature[47]. Research indicates that Snail represses Twist targets because their target sequences are very similar, and their binding is mutually

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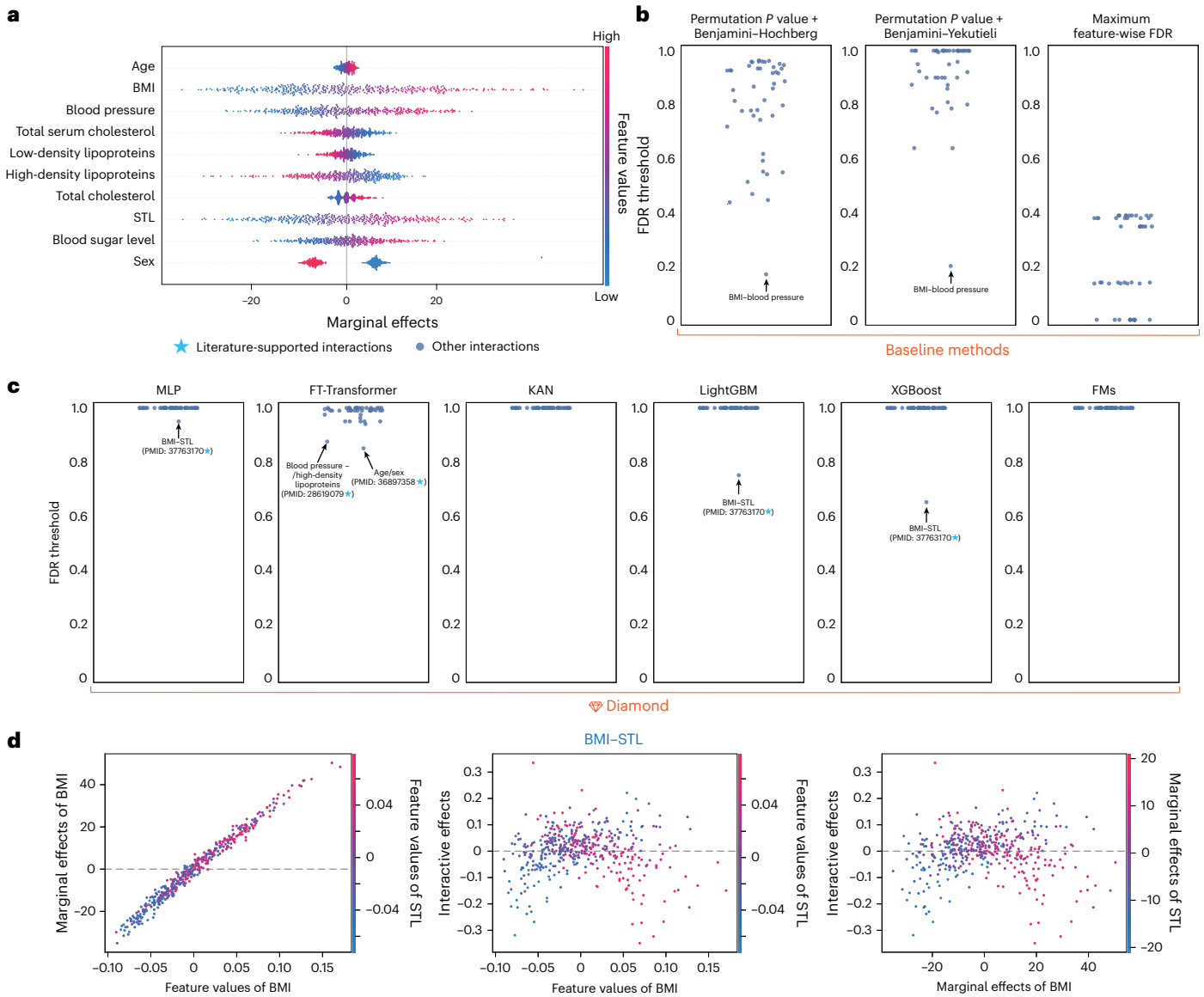


Fig. 4 | Evaluating Diamond on a real diabetes dataset. a, Each feature contributes differently to predicting disease progression, as measured by the Expected Gradient scores in the MLP model. **b**, Diamond is compared against three baseline methods. The blue stars indicate interactions supported by literature evidence, referenced by the accompanying PubMed identifiers. **c**, Diamond is used with various ML models to identify important non-additive interactions. Each possible interaction is measured by the minimum FDR threshold cut-off at which it is selected, with the top interaction annotated. **d**, Top interaction, between BMI and STL, is qualitatively evaluated from three aspects: the marginal importance measure, the interaction importance measure and the contribution of the marginal importance measures to the interaction importance measure.

exclusive. This literature evidence is further supported by a qualitative evaluation (Fig. 5d–f and Supplementary Figs. 4–6). Specifically, we examined the Snail–Twist interaction in terms of the contribution of feature values to the enhancer status prediction, observing that high Twist expression suppresses the contribution of Snail to enhancer activation.

Furthermore, we investigated the remaining identifications that lack supporting ground-truth or literature evidence. We discovered that these interactions can be logically explained through transitive effects. Specifically, even without a direct interaction between TF1 and TF3, we might still expect a strong interaction between them if solid interactions exist between TF1 and TF2 and between TF2 and TF3. On this basis, for example, the interaction between the TFs Twist (UniProt ID: P10627) and Zeste (UniProt ID: P09956), identified by Diamond+MLP, can be classified as a transitive interaction, supported by experimentally validated interactions[48]: (1) the interaction between Twist and Ultrabithorax (UniProt ID: P83949), Decapentaplegic (UniProt ID: P07713), Pho (UniProt ID: Q8ST83), Raf (UniProt ID: P11346), Psc (UniProt ID: P35820) and Trr (UniProt ID: Q8IRW8); and (2) the interaction between these proteins and Zeste. Analogously, the interaction between the TFs Twist (UniProt ID: P10627) and Dichaete (UniProt ID: Q24533), identified by Diamond together with two tree-based models, can also be supported by transitive interactions[48]: (1) the interaction between Twist and Pho, Raf, Psc, Trr, Myc (UniProt ID: Q9W4S7) and Trx (UniProt ID: P20659); and (2) the interaction between these proteins and Dichaete.

Finally, Diamond is able to identify non-additive interactions with various interactive patterns (Fig. 5d–f). Among the top five interactions reported by Diamond with MLP, the Zelda–Twist interaction, the Bicoid–Twist interaction and the Krueppel–Twist interaction exhibit synergistic effects. The high expression of both TFs in these interactions reinforces the contribution of each individual factor to enhancer activation. By contrast, the Snail–Twist interaction, which was

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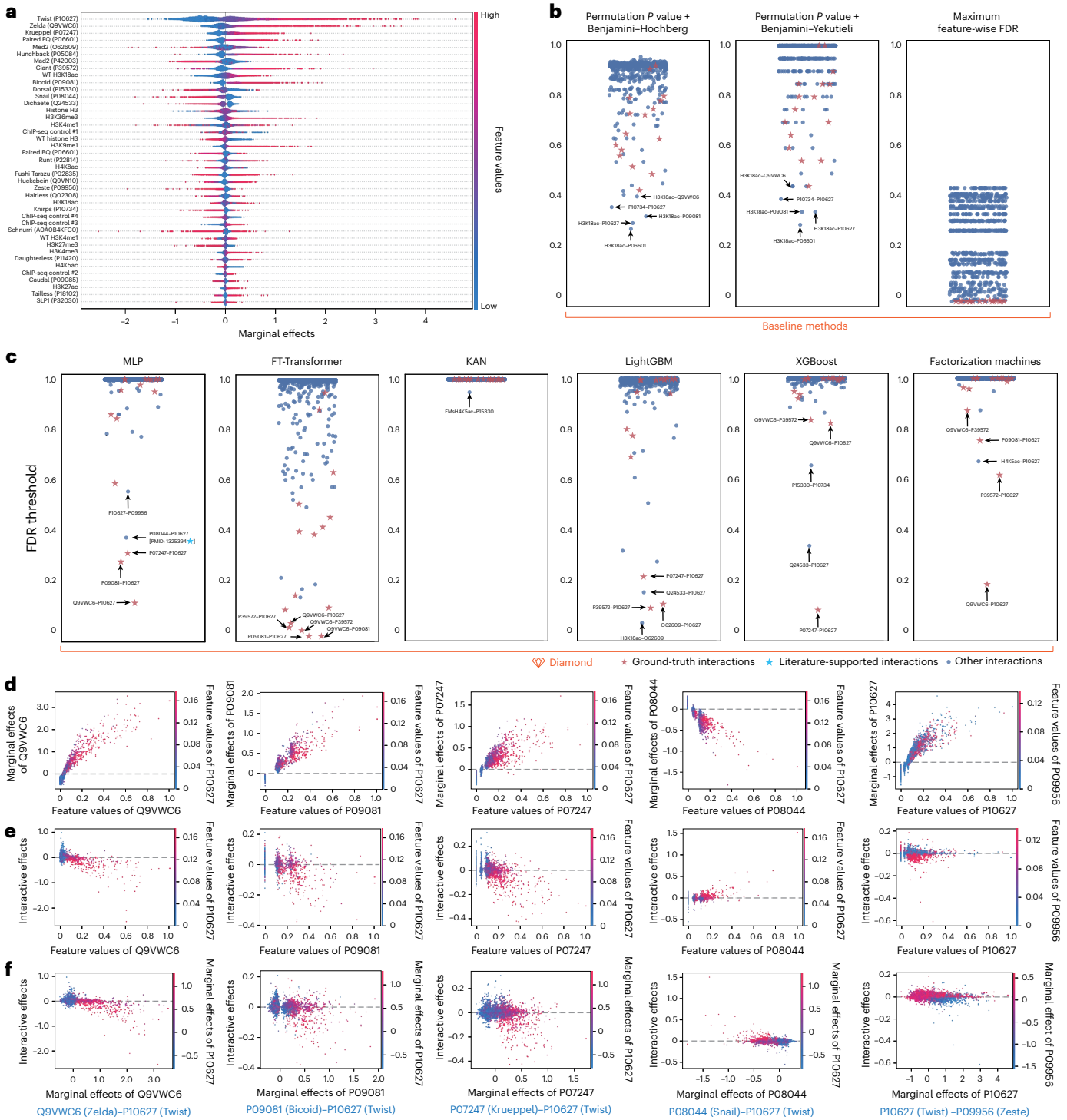


Fig. 5 | Evaluating Diamond on a real *Drosophila* enhancer dataset. a, Each feature contributes differently to predicting enhancer status, as measured by the Expected Gradient scores in the MLP model. **b**, Diamond is compared against three baseline methods. The annotated TFs are labelled by their UniProt identifiers. The red stars indicate well-characterized physical interactions in early *Drosophila* embryos as ground truth. The blue stars indicate interactions supported by literature evidence, referenced by the accompanying PubMed identifiers. **c**, Diamond is used with various ML models to identify important non-additive interactions. Each possible interaction is measured by the minimum FDR threshold cut-off at which it is selected, with the top five interactions annotated. The top interactions reported by the MLP model are qualitatively evaluated from three aspects: the contribution of feature values to the marginal (**d**) and interaction (**e**) importance measures, and the contribution of the marginal importance measures to the interaction importance measure (**f**).

mentioned earlier, exhibits repressive effects, where high expression of one TF suppresses the contribution of the other to enhancer activation. Last, the Twist–Zeste interaction exhibits alternative effects, where the high expression of either TF enhances the contribution of each to enhancer activation.

Diamond reveals drivers of health-related mortality

Last, we evaluated the performance of Diamond on a real mortality risk dataset[49] to investigate the relationship between mortality risk factors and long-term health outcomes within the US population. We used a mortality dataset from the National Health and Nutrition

https://doi.org/10.1038/s42256-025-01086-8

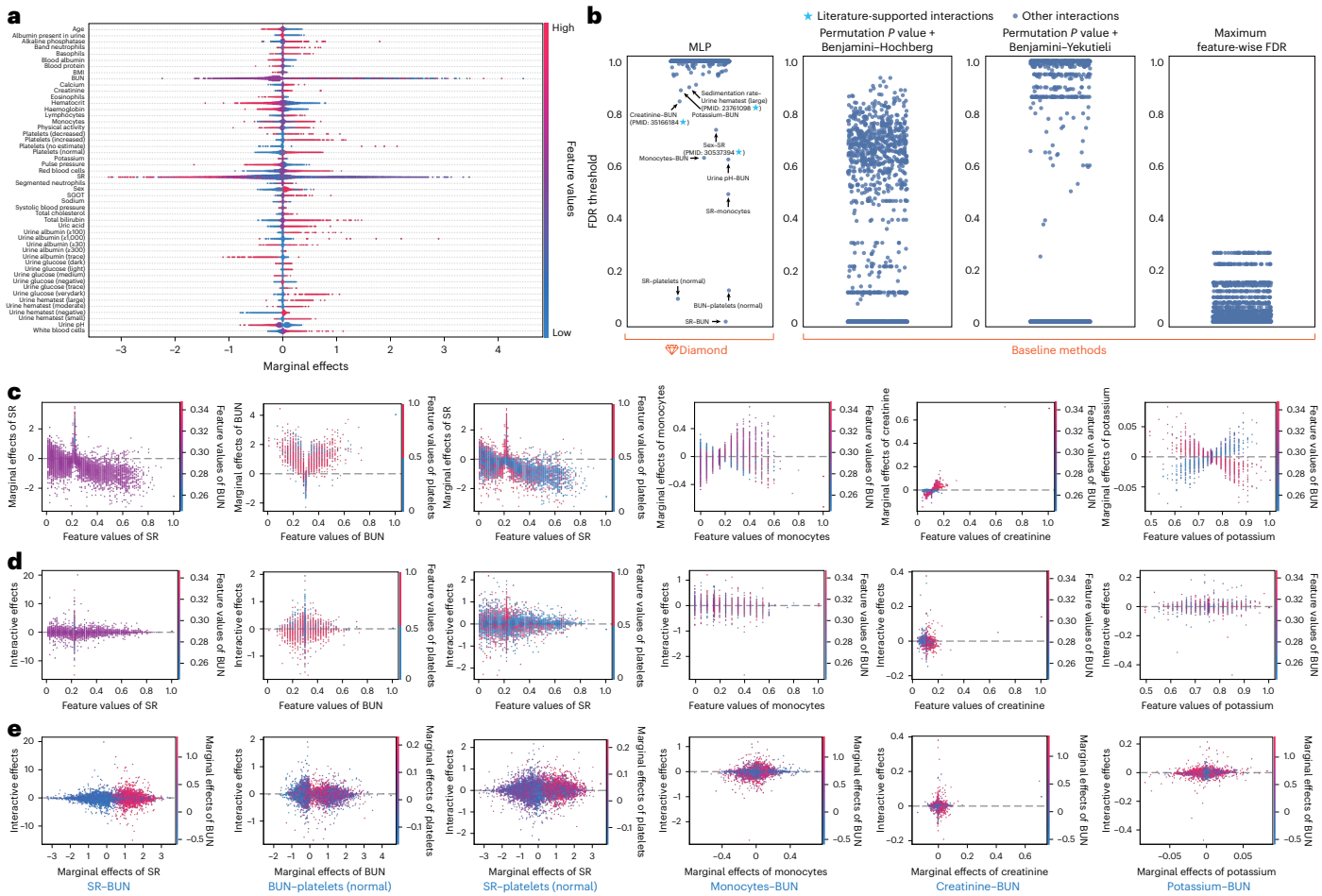


Fig. 6 | Evaluating Diamond on a real mortality risk dataset. a, Each feature contributes differently to predicting the mortality status, as measured by the Expected Gradient scores. It is worth mentioning that two marginally important features do not necessarily result in important interactions, as anticipated in Diamond's design. **b**, Diamond is used with the MLP model to identify important non-additive interactions and is compared against other baseline methods. Each possible interaction is measured by the minimum FDR threshold cut-off at which it is selected, with the top five interactions annotated. The blue stars indicate interactions supported by literature evidence, referenced by the accompanying PubMed identifiers. The survival loss function for predicting mortality restricts the modelling to DNNs, and we use MLP specifically due to memory limitations with other DNN models. The top interactions reported by Diamond are qualitatively evaluated from three aspects: the contribution of feature values to the marginal (**c**) and interaction (**d**) importance measures, and the contribution of the marginal importance measures to the interaction importance measure (**e**).

Examination Survey (NHANES I) and NHANES I Epidemiologic Follow-up Study (NHEFS). The dataset incorporates 35 clinical and laboratory measurements from 14,407 US participants between 1971 and 1974 (Fig. 6a). Note that some of the features are presented categorically, which makes model's trained from these data challenging to interpret. We addressed this problem by converting each categorical feature into a list of binary features using one-hot encoding. The dataset also reports the mortality status of participants as of 1992, with 4,785 individuals known to have passed away before 1992. The task is to construct an MLP-based model to predict the mortality status using a survival loss function. We assessed the trained model using fivefold cross-validation and selected the one with the best performance for model interpretation and interaction discovery. For robustness, we repeated the experiment 20 times using different random seeds. Each repetition involved knockoff generation, ML model training and interaction-wise FDR estimation.

Given the absence of ground-truth interactions for this task, we began by evaluating the interactions identified by Diamond through literature support. Our analysis shows that three of the top ten selected interactions are directly supported by existing literature (Fig. 6b). For example, we investigated the interaction between sex and sedimentation rate (SR) and found that although a high SR is strongly associated with a higher risk of overall mortality, this association can be attenuated by sex[50]. Additionally, we investigated the interaction between creatinine and the blood urea nitrogen (BUN) identified by Diamond. Research indicates that the BUN/creatinine ratio is recognized to have a nonlinear association with all-cause mortality and a linear association with cancer mortality[51]. The literature evidence is further supported by qualitative evaluation (Fig. 6c–e). Specifically, we examined the BUN–creatinine interaction in terms of the contribution of feature values to mortality status prediction, observing that when the creatinine level exceeds a certain level, a high BUN level further increases the mortality risk. By contrast, the baseline methods display significantly different behaviour, reporting an overwhelming number of interactions at the significant FDR level, which appears to overestimate false positives.

Finally, we probed into the remaining identifications that lack supporting literature evidence. As in the *Drosophila* analysis, we discovered that these interactions can be logically explained through transitive effects. For example, the interaction between BUN and potassium can be justified by combining the following two established facts: (1) BUN level is indicative of chronic kidney disease development[52], and (2) patients with chronic kidney diseases show a progressively increasing mortality rate with abnormal potassium levels[53]. Additionally, the interaction between BUN and SR reflects the fact that (1) the

creatinine level and SR are strongly associated to clinical pathology types and prognosis of patients[54], and (2) creatinine and BUN have a nonlinear association[51].

Discussion

In this study, we aim to enable rigorous data-driven scientific discoveries, which are crucial in the presence of massive datasets. For this purpose, we propose Diamond, an error-controlled interaction discovery method designed to work with a variety of ML models. Diamond comprises three key components. First, Diamond achieves FDR control using carefully designed knockoffs, without relying on p values, which are often difficult to obtain in generic ML models. Second, Diamond distills non-additive interaction effects, thereby maintaining FDR control at the target level. This non-additivity distillation step is critical because naive application of off-the-shelf feature interaction importance measures fail to control the FDR in this setting. Third, Diamond is versatile in discovering important non-additive interactions across a broad class of ML models, different knockoff designs and various interaction importance measures, simultaneously guaranteeing FDR control. We have applied Diamond to various simulated and real datasets to demonstrate its empirical utility, proving it to be the only valid solution for error-controlled data-driven scientific discovery compared with other alternative methods.

Methodologically, our approach provides a path towards enhancing transparency in complex ML models. The complexity of ML models has long posed a trade-off between predictability and interpretability. Specifically, sophisticated ML models such as DNNs excel at detecting subtle relationships and patterns within complex data. However, their black-box nature poses challenges in mission-critical domains in which error tolerance is minimal. By providing FDR control in this setting, we can better understand why and how models make predictions, thereby enabling more effective use of ML models to gain human-understandable insights. Diamond can help diagnose potential biases in ML models, contributing positively to the integrity, accountability and fairness required when using AI technologies.

Last, this study highlights several promising directions for future research. First, we observe that Diamond tends to be conservative. This suggests a future direction for improvement in enhancing statistical power through further refinement of the FDR estimation process. Previous efforts, like the knockoff-tailored plug-in pairwise-coupling layer in MLP models, have shown superior performance in statistical power[32]. However, these previously described methods are specific to certain types of ML model and may not be generally applicable to all generic ML models, indicating the need for improvement to broaden their applicability. Second, we observe that Diamond cannot distinguish between direct interactions and interactions caused by transitive effects. In reality, the former is scientifically more interesting than the latter, though both fit the definition of non-additive interactions. Specifically, even without a direct interaction between two features, we might still expect a strong interaction between them if solid interactions exist individually between each of these features and a third feature. Though transitive interactions are non-additive, for the purpose of scientific discovery, a potential research direction is to identify FDR-controlled direct interactions exclusively, using a causal inference framework[55]. Third, we observe that different ML models exhibit different levels of performance in identifying important interactions. Briefly, models with stronger modelling capabilities, such as transformer models, are better at identifying important interactions compared with models with weaker capabilities, such as FMs and convolutional neural networks (CNNs). As independent studies have highlighted, the sensitivity of interaction detection depends on the importance measures of the interactions, which are critically influenced by the quality of the model[56]. A potential research direction is to automatically determine the best ML models for interaction detection, rather than focusing solely on modelling accuracy. Finally, this study primarily focuses on discovering pairwise interactions. Although detecting pairwise interactions holds practical significance in various biological contexts, the discovery of higher-order interactions can naturally provide deeper insights for explaining genetic mechanisms, diseases and drug effects in healthcare domains. This problem is highly challenging in practice due to the exponentially large search space. A potential research direction for future studies is to effectively reduce the search space to generalize Diamond for practical discovery of higher-order interactions.

In conclusion, Diamond enables the error-controlled detection of non-additive feature interactions in the context of a variety of ML models. The versatility and flexibility of Diamond make it widely applicable in high-stakes and error-intolerant domains in which interpretability and statistical rigour are needed. The robustness against misspecified ML models and low-quality knockoffs ensures the reliability of Diamond, even when erroneously applied in extreme cases. We believe that this powerful tool will pave the way for the broader deployment of ML models in scientific discovery and hypothesis validation.

Methods

FDR control with knockoffs

Diamond achieves FDR control by leveraging the model-X knockoffs framework[26,27], which was proposed in the setting of error-controlled feature selection. The core idea of knockoffs is to generate dummy features that perfectly mimic the empirical dependence structure among the original features but are conditionally independent of the response given the original features. Briefly speaking, the knockoff filter achieves FDR control in two steps: (1) construction of knockoff features and (2) filtering using knockoff statistics.

For the first step, the knockoff features are defined as follows:

Definition 1. Model-X knockoff[27]: the model-X knockoff features for the family of random features $\mathbf{X} = (X_1 \ldots X_p)$ are a new family of random features $\tilde{\mathbf{X}} = (\tilde{X}_1, \ldots, \tilde{X}_p)$ that satisfy two properties:

1. $(\mathbf{X}, \tilde{\mathbf{X}})_{\text{swap}(\mathcal{S})} \overset{d}{=} (\mathbf{X}, \tilde{\mathbf{X}})$ for any subset $\mathcal{S} \subset \{1, \ldots, p\}$, where $\text{swap}(\mathcal{S})$ means swapping X_j and $\tilde{X}_j$ for each $j \in \mathcal{S}$ and $\overset{d}{=}$ denotes equal in distribution, and
2. $\tilde{\mathbf{X}} \perp\!\!\!\perp \mathbf{Y} | \mathbf{X}$, that is, $\tilde{\mathbf{X}}$ is independent of response $\mathbf{Y}$ given feature $\mathbf{X}$.

According to Definition 1, the construction of the knockoffs must be independent of the response $\mathbf{Y}$. Thus, if we can construct a set $\tilde{X}$ of model-X knockoff features properly, then by comparing the original features with these control features, FDR can be controlled at the target level q. In the Gaussian setting, that is, $\mathbf{X} \approx \mathcal{N}(0, \Sigma)$ with covariance matrix $\Sigma \in \mathbb{R}^{p \times p}$, the model-X knockoff features can be constructed easily:

$$\tilde{\mathbf{X}} | \mathbf{X} \approx N(\mathbf{X} - \text{diag}\{\mathbf{s}\}\Sigma^{-1}\mathbf{X}, 2\text{diag}\{\mathbf{s}\} - \text{diag}\{\mathbf{s}\}\Sigma^{-1}\text{diag}\{\mathbf{s}\}), \quad (2)$$

where $\text{diag}\{\mathbf{s}\}$ is a diagonal matrix with all components of $\mathbf{s}$ being positive such that the conditional covariance matrix in equation (2) is positive definite. As a result, the original features and the model-X knockoff features constructed by equation (2) have the following joint distribution.

$$(\mathbf{X}, \tilde{\mathbf{X}}) \approx \mathcal{N}\left(\begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \Sigma & \Sigma - \text{diag}\{\mathbf{s}\} \\ \Sigma - \text{diag}\{\mathbf{s}\} & \Sigma \end{pmatrix}\right) \quad (3)$$

With the constructed knockoff $\tilde{\mathbf{X}}$, feature importances are quantified by computing the knockoff statistics $W_j = g_j(Z_j, \tilde{Z}_j)$ for $1 \le j \le p$, where Z_j and $\tilde{Z}_j$ represent feature importance measures for the jth feature X_j and its knockoff counterpart $\tilde{X}_j$, respectively, and $g_j(\cdot, \cdot)$ is an antisymmetric function satisfying $g_j(Z_j, \tilde{Z}_j) = -g_j(\tilde{Z}_j, Z_j)$. The knockoff statistics W_j should satisfy a coin-flip property such that swapping an arbitrary pair X_j and its knockoff counterpart $\tilde{X}_j$ only changes the sign

of W_j but keeps the signs of other W_k ($k \neq j$) unchanged[27]. A desirable property for knockoff statistics W_j is that important features are expected to have large absolute values, whereas unimportant ones should have small symmetric values around 0.

Finally, the absolute values of the knockoff statistics $|W_j|$ are sorted in decreasing order, and FDR-controlled features are those whose W_j values exceed some threshold T. In particular, the choice of threshold T follows $T = \min\{t \in \mathcal{W}, \frac{1+|\{j:W_j \leq -t\}|}{|\{j:W_j \geq t\}|} \leq q\}$, where $\mathcal{W} = \{|W_j| : 1 \leq j \leq p\} \setminus \{0\}$ is the set of unique non-zero values from $|W_j|$ values and $q \in (0,1)$ is the desired FDR level specified by the user.

Measuring the non-additive interaction effect

As a key precursor to FDR estimation, Diamond quantifies feature interactions from trained ML models and produces a ranked list of these interactions, with higher-ranked interactions indicating greater importance. For notational simplicity, we use indices for both original features and knockoffs as $\{1, 2 \ldots 2p\}$, with $\{1 \ldots p\}$ and $\{p+1 \ldots 2p\}$ corresponding to the original features and their respective knockoff counterparts. Here we define $\mathbf{E}^{2D} = [e_{ij}]_{i,j=1}^{2p} \in \mathbb{R}^{2p \times 2p}$ as a reported interaction importance measure from existing methods. There are many feature interaction importance measures available for $\mathbf{E}^{2D}$, each attributing the prediction influence to feature pairs in different ways. However, it is important to note that such measures favour pairs in which both features are simultaneously important for a model's prediction, rather than capturing the true non-additive interaction effects between the two features[21]. Further supported by simulation studies (Fig. 2c), we observed that many feature interaction importance measures tend to assign higher interaction scores to two marginally important but non-interacting features compared with two random ones, even though neither pair has a real interaction, leading to the failure of FDR control.

The direct reason for the interaction importance measure falling short in controlling FDR is its violation of the knockoff filter's assumption. Specifically, the knockoff filter requires that the importance scores of knockoff-involving interactions and false interactions have a similar distribution. To resolve this issue, we introduce a non-additivity distillation procedure to be applied on top of existing interaction importance measures. Specifically, we consider that a reported interaction importance measure from existing methods comprises a mixture of several factors: label-dependent marginal effects for individual features, label-independent feature biases, independent random noise and potential non-additive interaction effects between feature pairs. Thus, the reported interaction between features i and j is represented as

$$e_{ij} = s_{ij} + g_i(e_i) + g_j(e_j) + b(I_{ij}) + \varepsilon_{ij}, \tag{4}$$

where $\varepsilon_{ij} \in \mathbb{R}$ is random noise independent of both features and predictions and $e_{ij} \in \mathbb{R}$ and $e_i, e_j \in \mathbb{R}$ are reported pairwise and univariate feature importance measures that are dependent on the model's predictions, respectively. The functions $g_i, g_j : \mathbb{R} \mapsto \mathbb{R}$ adapt univariate feature importance to be compatible with feature interaction importance. The function $b : \mathbb{R}^{2p} \mapsto \mathbb{R}$ models the feature-specific biases that are independent of the model's predictions, where $I_{ij} \in \{0,1\}^{2p}$ indicates the presence of features i and j. Our goal is to identify s_{ij}, the potential non-additive interaction effects between features i and j.

We formulate the identification of interaction effects s_{ij} as the residuals of a regression task:

$$\min_{b,g_1,g_2,\ldots} \sum_{i<j} w_{ij} \cdot \left\| e_{ij} - g_i(e_i) - g_j(e_j) - b(I_{ij}) \right\|^2, \tag{5}$$

where $w_{ij} > 0$ is the conditional probability of being either original-only (that is, $i,j \leq p$) or knockoff-involving (that is, $i > p$ or $j > p$) feature pairs given two univariate feature importance measures e_i and e_j, estimated by a logistic regression model[57]. The rationale is based on the important observation that most feature pairs do not exhibit non-additive interactions, especially those involving knockoff features. Therefore, we want to focus more on potential non-additive interactions that have large univariate feature importance, as important interactions naturally consist of significant marginal features. In this study, we parameterize the functions $b : \mathbb{R}^{2p} \mapsto \mathbb{R}$ and $g_i, g_j : \mathbb{R} \mapsto \mathbb{R}$ using generalized additive models and optimize equation (5) using the py GAM library[58].

FDR control for interactions

After calculating the non-additive interaction effects using equation (5), we denote the resultant set of interaction effects as $\Gamma = \{s_{ij} | i < j, i \neq j - p\}$. We arrange Γ in decreasing order and select interactions for which the interaction effect, Γ_j, exceeds some threshold, T. This selection ensures that the chosen interactions adhere to a desired FDR level $q \in (0,1)$.

However, a point of complexity is introduced due to the heterogeneous interactions, which include original-only and knockoff-involving interactions. The latter further comprises original–knockoff, knockoff–original and knockoff–knockoff interactions. Following the strategy outlined in ref. 59, the threshold T is determined by

$$T = \min\left\{ t \in \mathcal{T}, \ \frac{|\{j : \Gamma_j \geq t, j \in \mathcal{K}\}| - 2 \cdot |\{j : \Gamma_j \geq t, j \in \mathcal{KK}\}|}{|\{j : \Gamma_j \geq t, j \notin \mathcal{K} \text{ and } j \notin \mathcal{KK}\}|} \leq q \right\}, \tag{6}$$

where $\mathcal{K}$ and $\mathcal{KK}$ denote the sets of interactions that include at least one knockoff feature and both knockoff features, respectively, whereas $\mathcal{T}$ refers to the set of unique non-zero values present in Γ. Supplementary Section 4 provides a more detailed description of FDR estimation.

ML models

Diamond is designed to be compatible with a broad class of ML models. To demonstrate the broad applicability of Diamond, we use it in conjunction with ML models from methodologically distinct categories, including DNN models, KANs[28], tree-based models and factorization-based models.

For DNNs, we selected three representative models: MLP, CNN and transformer models. It is worth noting that CNNs are not well suited for modelling tabular data and are included here as an example of model misspecification. We configured a pyramid-shaped MLP model with the exponential linear unit activation and four hidden layers, each having neuron sizes of $2p$, p, $p/2$, and $p/4$, respectively, where p denotes the input dimensionality. We configured a CNN model consisting of two convolutional layers. The first layer has one input channel and 32 output channels, whereas the second layer has 32 input channels and 64 output channels, followed by max pooling. Batch normalization and rectified linear unit activation are applied after each convolution. The output is then flattened and passed through two fully connected layers, mapping to 128 units before reaching the output dimension. For transformer models, we used FT-Transformer[29], a model specifically designed for tabular data. By default, it consists of six layers, eight attention heads, a 32-dimensional feed-forward network and a 0.1 dropout rate. All DNN models follow the guidance of ref. 32 and include a plug-in pairwise-coupling layer, connecting each original feature with its knockoff counterpart in a pairwise fashion to maximize statistical power (Supplementary Section 1).

For KANs, we configured a two-layer model, with a hidden layer of width $p/2$, followed by a mapping to the output dimension. For tree-based models, we used two widely used gradient-boosted decision tree representatives—XGBoost[60] and LightGBM[61]. We used the implementations provided by the two libraries with default settings. For factorization-based models, we used a widely adopted representative

method—FM[62]. We used 2-Way FM implemented in the xLearn library[63] with default settings, which models up to second-order feature interactions.

Model interpretation

Diamond utilizes representative importance measures to interpret the trained ML models and elucidate the relationship between features and responses. For DNN and KAN models, we used a representative model-agnostic interaction importance measure, Expected Hessian[36], to clarify the relationships between features and responses without making assumptions about any specific model architecture (Supplementary Section 3). To demonstrate Diamond's flexibility, we also used Integrated Hessian[16], another state-of-the-art, model-agnostic interaction importance measure, alongside a model-specific interaction importance measure derived from the model weights, specifically designed for the MLP architecture (Supplementary Section 2). Both model-agnostic interaction importance measures, Expected Hessian and Integrated Hessian, are computed using the Path-Explain library[16]. For tree-based models such as XGBoost and LightGBM, we used TreeSHAP[15], a Shapley-value-based interpretation method specifically designed for tree-based ML models, to clarify the relationships between features and responses. We used the implementation provided by the SHAP library[7] to calculate both univariate and pairwise feature importance values. For factorization-based models, since 2-Way FM is explicitly modelled as the weighted sum of univariate and pairwise feature interactions, we used the learned coefficients of these interactions as the corresponding importance measures.

Knockoff generation

Diamond relies on knockoffs to control the FDR. It is worth mentioning that conventional knockoffs are limited to Gaussian settings, which may not be applicable in many practical scenarios. In this study, we focused on the state-of-the-art knockoff design without assuming any specific feature distribution. In particular, we considered commonly used non-Gaussian knockoff generation methods such as KnockoffsDiagnostics[31], KnockoffGAN[33], Deep knockoffs[34] and VAE knockoffs[35]. Furthermore, to assess Diamond's robustness in the presence of poorly generated knockoffs, we created invalid knockoffs by independently permuting each feature across samples, thereby violating the definition of knockoffs outlined in the 'FDR control with knockoffs' section.

Alternative FDR estimation methods

We evaluate the performance of Diamond compared with three baseline methods. For the first two baseline methods, we use a permutation-based approach to calculate the interaction-wise FDR. Specifically, this involves using a previously described permutation procedure tailored for neural networks to assess the significance of interactions and calculate the permutation p values[64]. The permutation p values are combined with the Benjamini–Hochberg procedure[25] or the Benjamini–Yekutieli procedure[65] to estimate the FDR. The difference between the Benjamini–Hochberg and Benjamini–Yekutieli procedures is that the former assumes the p values are independent or positively dependent, whereas the latter imposes no assumptions on the dependencies among p values.

For the second baseline method, we consider an ensemble-based approach that represents interaction-wise FDR as the aggregation of feature-wise FDR. This approach follows the intuition that an important feature interaction is composed of important univariate features. Specifically, we use a previously described knockoff-based procedure tailored for neural networks to estimate the feature-wise FDR of each univariate feature[32]. We then approximate the interaction-wise FDR as the maximum of the two comprising univariate feature-wise FDRs.

Data availability

All public datasets used for evaluating the models are available via GitHub at https://github.com/batmen-lab/diamond/blob/main/data.zip and via Zenodo at https://doi.org/10.5281/zenodo.15733825 (ref. 66).

Code availability

The software implementation and all models described in this study are available via GitHub at https://github.com/batmen-lab/diamond and via Zenodo at https://doi.org/10.5281/zenodo.15733825 (ref. 66).

References

1. Obermeyer, Z. & Emanuel, E. J. Predicting the future–big data, machine learning, and clinical medicine.*N. Engl. J. Med.* **375**, 1216–1219 (2016).
2. Lipton, Z. C. The mythos of model interpretability: in machine learning, the concept of interpretability is both important and slippery. *Queue* **16**, 31–57 (2018).
3. Agrawal, A., McHale, J. & Oettl, A. Artificial intelligence and scientific discovery: a model of prioritized search. *Res. Policy* **53**, 104989 (2024).
4. Samek, W., Montavon, G., Lapuschkin, S., Anders, C. J. & Müller, K.-R. Explaining deep neural networks and beyond: a review of methods and applications. *Proc. IEEE* **109**, 247–278 (2021).
5. Simonyan, K., Vedaldi, A. & Zisserman, A. Deep inside convolutional networks: visualising image classification models and saliency maps. In *2nd International Conference on Learning Representations, Workshop Track* (eds Bengio, Y. & LeCun, Y.) (2014).
6. Shrikumar, A., Greenside, P., Shcherbina, A. & Kundaje, A. Learning important features through propagating activation differences. In *Proc. 34th International Conference on Machine Learning* 3145–3153 (PMLR, 2017).
7. Lundberg, S. M. & Lee, S.-I. A unified approach to interpreting model predictions. *Adv. Neural Inf. Process. Syst.* **30**, 4765–4774 (2017).
8. Sundararajan, M., Taly, A. & Yan, Q. Axiomatic attribution for deep networks. In *Proc. 34th International Conference on Machine Learning* 3319–3328 (PMLR, 2017).
9. Lu, Y. Y., Guo, W., Xing, X. & Noble, W. S. DANCE: enhancing saliency maps using decoys. In *Proc. 38th International Conference on Machine Learning* 7124–7133 (PMLR, 2021).
10. Tsang, M., Cheng, D. & Liu, Y. Detecting statistical interactions from neural network weights. In *Proc. International Conference on Learning Representations* (ICLR, 2018).
11. Lu, Y. Y. & Noble, W. S. A wider field of view to predict expression. *Nat. Methods* **18**, 1155–1156 (2021).
12. Watson, D. S. Interpretable machine learning for genomics. *Hum. Genet.* **141**, 1499–1513 (2022).
13. Tsang, M., Liu, H., Purushotham, S., Murali, P. & Liu, Y. Neural interaction transparency (NIT): disentangling learned interactions for improved interpretability. *Adv. Neural Inf. Process. Syst.* **31**, 5809–5818 (2018).
14. Cui, T., Marttinen, P. & Kaski, S. Learning global pairwise interactions with Bayesian neural networks. In *Proc. 24th European Conference on Artificial Intelligence* (ECAI, 2020)
15. Lundberg, S. M. et al. From local explanations to global understanding with explainable AI for trees. *Nat. Mach. Intell.* **2**, 56–67 (2020).
16. Janizek, J. D., Sturmfels, P. & Lee, S.-I. Explaining explanations: axiomatic feature interactions for deep networks. *J. Mach. Learn. Res.* **22**, 104:1–104:54 (2021).
17. Sundararajan, M., Dhamdhere, K. & Agarwal, A. The Shapley Taylor interaction index. In *Proc. 37th International Conference on Machine Learning* 9259–9268 (PMLR, 2020).

18. Chang, C., Caruana, R. & Goldenberg, A. NODE-GAM: neural generalized additive model for interpretable deep learning. In *Proc. International Conference on Learning Representations* (ICLR, 2022).
19. S. Lerman, C. Venuto, H. Kautz & C. Xu. Explaining local, global, and higher-order interactions in deep learning. In *Proc. IEEE/CVF International Conference on Computer Vision* 1224–1233 (IEEE, 2021).
20. Zhang, H., Xie, Y., Zheng, L., Zhang, D. & Zhang, Q. Interpreting multivariate Shapley interactions in DNNs. In *Proc. Thirty-Fifth AAAI Conference on Artificial Intelligence* **35**, 10877–10886 (AAAI Press, 2021).
21. Tsang, M., Enouen, J. & Liu, Y. Interpretable artificial intelligence through the lens of feature interaction. Preprint at https://arxiv.org/abs/2103.03103 (2021).
22. Ghorbani, A., Abid, A. & Zou, J. Interpretation of neural networks is fragile. In *Proc. Thirty-Third AAAI Conference on Artificial Intelligence* 3681–3688 (AAAI Press, 2019).
23. Kindermans, P.-J. et al. in *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning* 267–280 (Springer, 2019).
24. Lu, Y. Y., Yu, T. C., Bonora, G. & Noble, W. S. ACE: explaining cluster from an adversarial perspective. In *Proc. 38th International Conference on Machine Learning* 7156–7167 (PMLR, 2021).
25. Benjamini, Y. & Hochberg, Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. Ser. B* **57**, 289–300 (1995).
26. Barber, R. F. & Candès, E. J. Controlling the false discovery rate via knockoffs. *Ann. Stat.* **43**, 2055–2085 (2015).
27. Candès, E. J., Fan, Y., Janson, L. & Lv, J. Panning for gold: model-X knockoffs for high-dimensional controlled variable selection. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **80**, 551–577 (2018).
28. Liu, Z. et al. KAN: Kolmogorov-Arnold networks. In *Proc. International Conference on Learning Representations* (eds Yue, Y. et al.) 70367–70413 (ICLR, 2025).
29. Gorishniy, Y., Rubachev, I., Khrulkov, V. & Babenko, A. Revisiting deep learning models for tabular data. *Adv. Neural Inf. Process. Syst.* **34**, 18932–18943 (2021).
30. Sorokina, D., Caruana, R., Riedewald, M. & Fink, D. Detecting statistical interactions with additive groves of trees. In *Proc. 25th International Conference on Machine Learning* 1000–1007 (ACM, 2008).
31. Blain, A., Thirion, B., Linhart, J. & Neuvial, P. When knockoffs fail: diagnosing and fixing non-exchangeability of knockoffs. Preprint at https://arxiv.org/abs/2407.06892 (2024).
32. Lu, Y. Y., Fan, Y., Lv, J. & Noble, W. S. DeepPINK: reproducible feature selection in deep neural networks. *Adv. Neural Inf. Process. Syst.* **31**, 8690–8700 (2018).
33. Tsang, M., Yoon, J. & van der Schaar, M. KnockoffGAN: generating knockoffs for feature selection using generative adversarial networks. In *Proc. International Conference on Learning Representations* (ICLR, 2019).
34. Romano, Y., Sesia, M. & Candès, E. Deep knockoffs. *J. Am. Stat. Assoc.* **115**, 1861–1872 (2020).
35. Liu, Y. & Zheng, C. Auto-encoding knockoff generator for FDR controlled variable selection. Preprint at https://arxiv.org/abs/1809.10765 (2018).
36. Erion, G., Janizek, J. D., Sturmfels, P., Lundberg, S. M. & Lee, S.-I. Improving performance of deep learning models with axiomatic attribution priors and expected gradients. *Nat. Mach. Intell.* **3**, 620–631 (2021).
37. Efron, B., Hastie, T., Johnstone, I. & Tibshirani, R. Least angle regression. *Ann. Stat.* **32**, 407–499 (2004).
38. Garbuzova, E. V. et al. Triglycerides, obesity and education status are associated with the risk of developing type 2 Diabetes in young adults, cohort study. *J. Pers. Med.* **13**, 1403 (2023).
39. Zhao, H. et al. High body mass index and triglycerides help protect against osteoporosis in patients with type 2 diabetes mellitus. *J. Diabetes Res.* **2020**, 1517879 (2020).
40. Huang, X. et al. Lower baseline serum triglyceride levels are associated with higher decrease in body mass index after laparoscopy sleeve gastrectomy among obese patients. *Front. Endocrinol.* **12**, 633856 (2021).
41. Shimizu, Y. et al. Association between high-density lipoprotein-cholesterol and hypertension in relation to circulating CD34-positive cell levels. *J. Physiol. Anthropol.* **36**, 26 (2017).
42. Kautzky-Willer, A., Leutner, M. & Harreiter, J. Sex differences in type 2 diabetes. *Diabetologia* **66**, 986–1002 (2023).
43. Basu, S., Kumbier, K., Brown, J. B. & Yu, B. Iterative random forests to discover predictive and stable high-order interactions. *Proc. Natl Acad. Sci. USA* **115**, 1943–1948 (2018).
44. Li, X.-Y. et al. Transcription factors bind thousands of active and inactive regions in the *Drosophila* blastoderm. *PLoS Biol.* **6**, e27 (2008).
45. Rivera-Pomar, R. & Jackle, H. From gradients to stripes in *Drosophila* embryogenesis: filling in the gaps. *Trends Genet.* **12**, 478–483 (1996).
46. Liska, O. et al. TFLink: an integrated gateway to access transcription factor–target gene interactions for multiple species. *Database* **2022**, baac083 (2022).
47. Ip, Y. T., Park, R. E., Kosman, D., Bier, E. & Levinec, M. The dorsal gradient morphogen regulates stripes of rhomboid expression in the presumptive neuroectoderm of the *Drosophila* embryo. *Genes Dev.* **6**, 1728–1739 (1992).
48. Yevshin, I., Sharipov, R., Valeev, T., Kel, A. & Kolpakov, F. GTRD: a database of transcription factor binding sites identified by ChIP-seq experiments. *Nucleic Acids Res.* **45**, D61–D67 (2017).
49. Cox, C. S. et al. Plan and operation of the NHANES I Epidemiologic Followup Study, 1992. *Vital Health Stat.* **35**, 1–231 (1997).
50. Fest, J. et al. Erythrocyte sedimentation rate as an independent prognostic marker for mortality: a prospective population-based cohort study. *J. Intern. Med.* **285**, 341–348 (2019).
51. Shen, S., Yan, X. & Xu, B. The blood urea nitrogen/creatinine (BUN/cre) ratio was U-shaped associated with all-cause mortality in general population. *Ren. Fail.* **44**, 184–190 (2022).
52. Collins, A. J. et al. Association of serum potassium with all-cause mortality in patients with and without heart failure, chronic kidney disease, and/or diabetes. *Am. J. Nephrol.* **46**, 213–221 (2017).
53. Seki, M. et al. Blood urea nitrogen is independently associated with renal outcomes in Japanese patients with stage 3–5 chronic kidney disease: a prospective observational study. *BMC Nephrol.* **20**, 115 (2019).
54. Liang, H. et al. Serum creatinine level and ESR values associated to clinical pathology types and prognosis of patients with renal injury caused by ANCA-associated vasculitis. *Exp. Ther. Med.* **14**, 6059–6063 (2017).
55. Luo, Y., Peng, J. & Ma, J. When causal inference meets deep learning. *Nat. Mach. Intell.* **2**, 426–427 (2020).
56. Adebayo, J. et al. Sanity checks for saliency maps. *Adv. Neural Inf. Process. Syst.* **31**, 9505–9515 (2018).
57. Freedman, D. A. & Berk, R. A. Weighting regressions by propensity scores. *Eval. Rev.* **32**, 392–409 (2008).
58. Serven, D. & Brummitt, C. pyGAM: generalized additive models in Python. *Zenodo* https://doi.org/10.5281/zenodo.1208723 (2018).
59. Walzthoeni, T. et al. False discovery rate estimation for cross-linked peptides identified by mass spectrometry. *Nat. Methods* **9**, 901–903 (2012).
60. Chen, T. & Guestrin, C. XGBoost: a scalable tree boosting system. In *Proc. 22nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining* 785–794 (ACM, 2016).

61. Ke, G. et al. LightGBM: a highly efficient gradient boosting decision tree. *Adv. Neural Inf. Process. Syst.* **30**, 3146–3154 (2017).
62. Rendle, S. Factorization machines. In *Proc. IEEE International Conference on Data Mining* 995–1000 (IEEE, 2010).
63. Ma, C. xLearn Python library (2019); https://xlearn-doc.readthedocs.io/
64. Cui, T. et al. Gene–gene interaction detection with deep learning. *Commun. Biol.* **5**, 1238 (2022).
65. Benjamini, Y. & Yekutieli, D. The control of the false discovery rate in multiple testing under dependency. *Ann. Stat.* **29**, 1165–1188 (2001).
66. Jiang, Y. et al. batmen-lab/diamond: the first release of diamond. *Zenodo* https://doi.org/10.5281/zenodo.15733825 (2025).

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Author contributions

W.C. and Y.J. implemented the code and performed the analysis. W.C. and Y.J. set up and preprocessed the datasets. W.C., Y.J. and Y.Y.L. prepared the figures. Y.Y.L. wrote the manuscript. All authors participated the discussion. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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